KNOWLEDGE AREA 1: INFORMATION TECHNOLOGY

IT. 412: Information Management

LEARNING OBJECTIVES

- Define Information Technology and its nature
- Analyze managerial issues related to Information Technology
- Evaluate the impact of Information Technology on organizations
- Define and distinguish between architecture and infrastructure
- Conduct the steps in transforming strategic goals into IT infrastructure
- Understand the structure of enterprise architectures

WHAT IS INFORMATION TECHNOLOGY?



WHAT IS INFORMATION TECHNOLOGY?

- Information technology refers to specific technical components:
  - Hardware
  - Software
  - Communications
- IT components alone may have no useful function and must be engineered into usable systems that serve organizational needs.

WHAT IS INFORMATION TECHNOLOGY?

Table 3.1 Some examples of "traditional" technology types

Hardware	Software	Communications
Central processing unit	Operating system	Modem
Disk drive	Job scheduler	Multiplexer
Printer	I/O controller	Networking switch
Keyboard	Program code compiler	Terminal
Magnetic tape drive	Device driver	Communications line
	Database system	Optical fibre

NATURE OF TECHNOLOGY

- The traditional categories of IT are becoming less relevant as devices like smartphones integrate thousands of components into a single unit.
- It is more useful to **focus on enduring trends and patterns of innovation** and obsolescence rather than on specific, rapidly changing technological miracles.
- Some experts argue that IT is now **highly commoditized and pre-packaged**, allowing for easy, almost effortless purchase and integration from a catalogue.
- The real, time-consuming effort now lies in **loading the specific content and essential details required** to make a system functional for a business, which requires significant human endeavor.

MANAGERIAL ISSUES IN IT

- **Technology management scope** - Increasingly outsourced technology requires senior management to weigh risks and benefits to determine investments in hardware, software, and trained personnel.
- **Managing suppliers** - Organizations must identify and cultivate the most useful relationship type with technology suppliers while recognizing different relationship kinds and managing costs. In most cases, being a good customer sometimes outweighs aggressive supplier treatment.
- **Acquiring technology** - Effective acquisition management requires careful justification and tendering procedures to avoid excessive "just in case" spending, alongside clear criteria for selecting appropriate supply arrangements under varying circumstances.

MANAGERIAL ISSUES IN IT

- **Managing the technology portfolio** - A portfolio perspective aligns current and future technology investments with business strategies, ensuring awareness of owned technology and its physical location.
- **Technology competencies** - Even with heavy supplier reliance, every organization must maintain internal competencies to track developments, understand IT's nature and origins, and foster experimentation, innovation, and evaluation.
- **Budget management** - A single budget is rarely optimal for the acquisition of new technology. Therefore, policies should support both centralized capital budgets and local departmental funding, with clear understanding and application of these approaches to be revisited later.

WHAT IS IT ARCHITECTURE? IT INFRASTRUCTURE?



ARCHITECTURE VS INFRASTRUCTURE

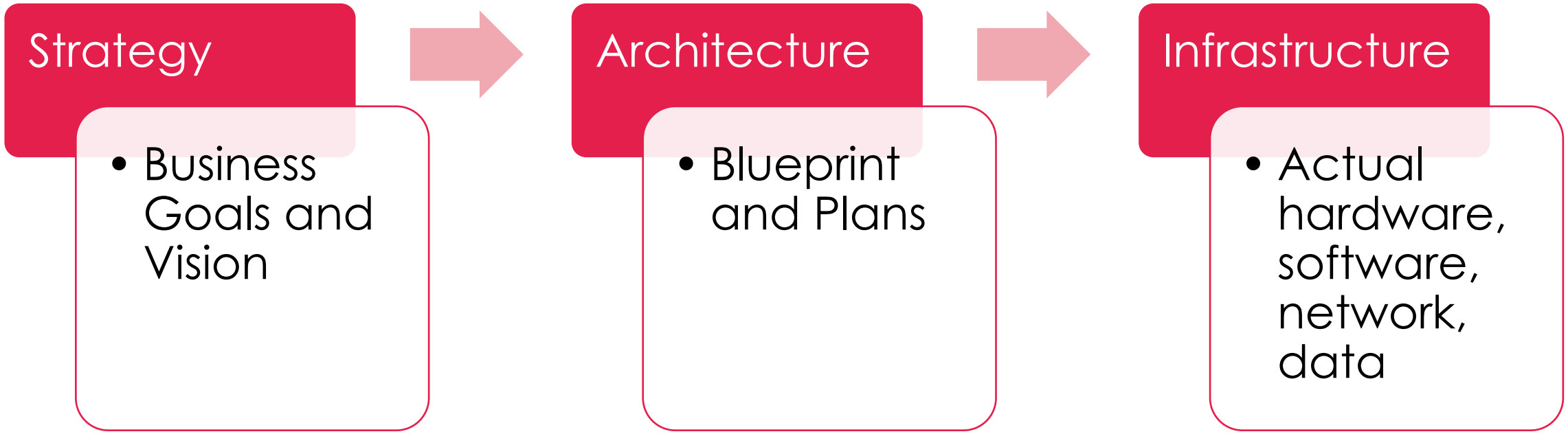
IT Architecture

- Blueprint for translating business strategy into a system plan
- Defines what needs to be built

IT Infrastructure

- Everything that supports the flow and processing of information
- Includes hardware, software, data, and network components that are chosen and assembled
- Defines what is built

THE BUSINESS STRATEGY TO IT INFRASTRUCTURE PROCESS



WHAT IS THE MANAGER'S ROLE?

- Communicate business vision to IT architects
- Understand what to expect from architecture and infrastructure
- Ensure plans don't block important business considerations
- Be a knowledgeable participant in discussions

FROM BUSINESS STRATEGY TO IT ARCHITECTURE

- The **strategy** defines the overarching direction for a business
- The **specific goals** develop measurable goals in line with the strategy
- The **business requirements** list down what the system must accomplish
- The **architectural requirements** identify the data, process, and security needs
- All culminate into the **architecture** which is the designed blueprint.

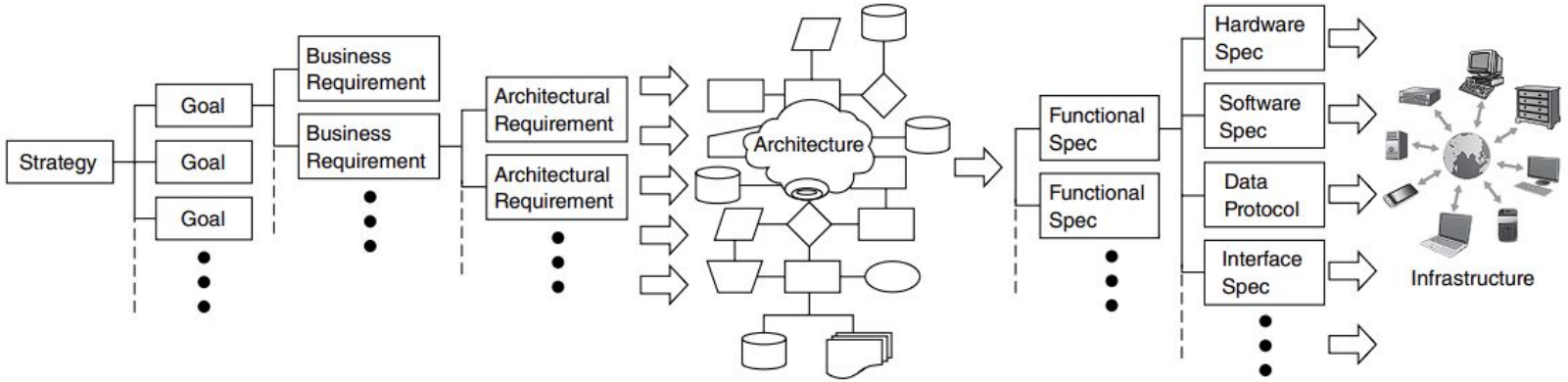


FIGURE 6.2 From strategy to architecture to infrastructure.

FROM IT ARCHITECTURE TO INFRASTRUCTURE

- The **functional specifications** describe what each component must do
- The **technical specifications** detail how to implement the system
- The **components** must be combined in a coherent structure to create a viable infrastructure

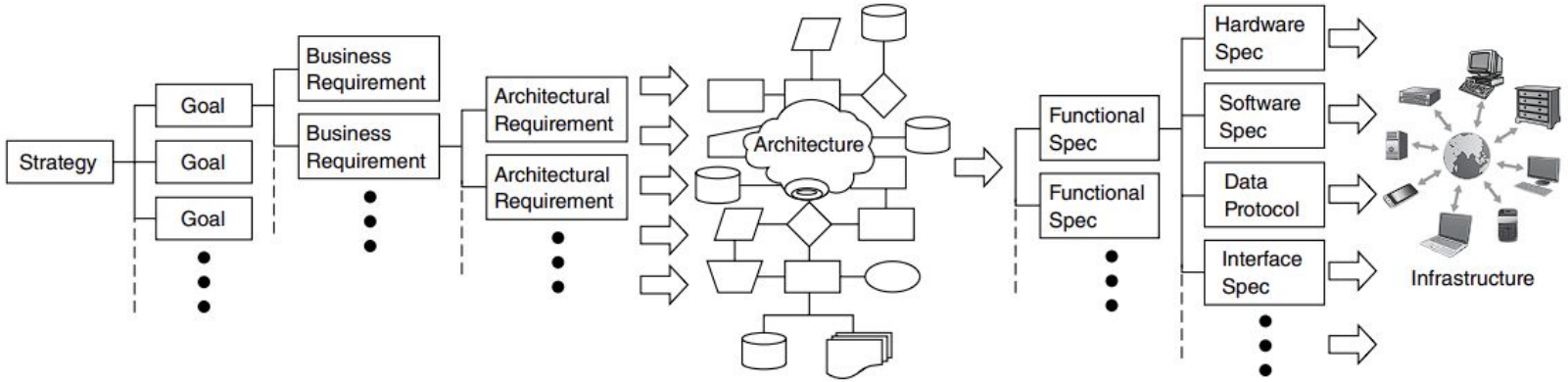
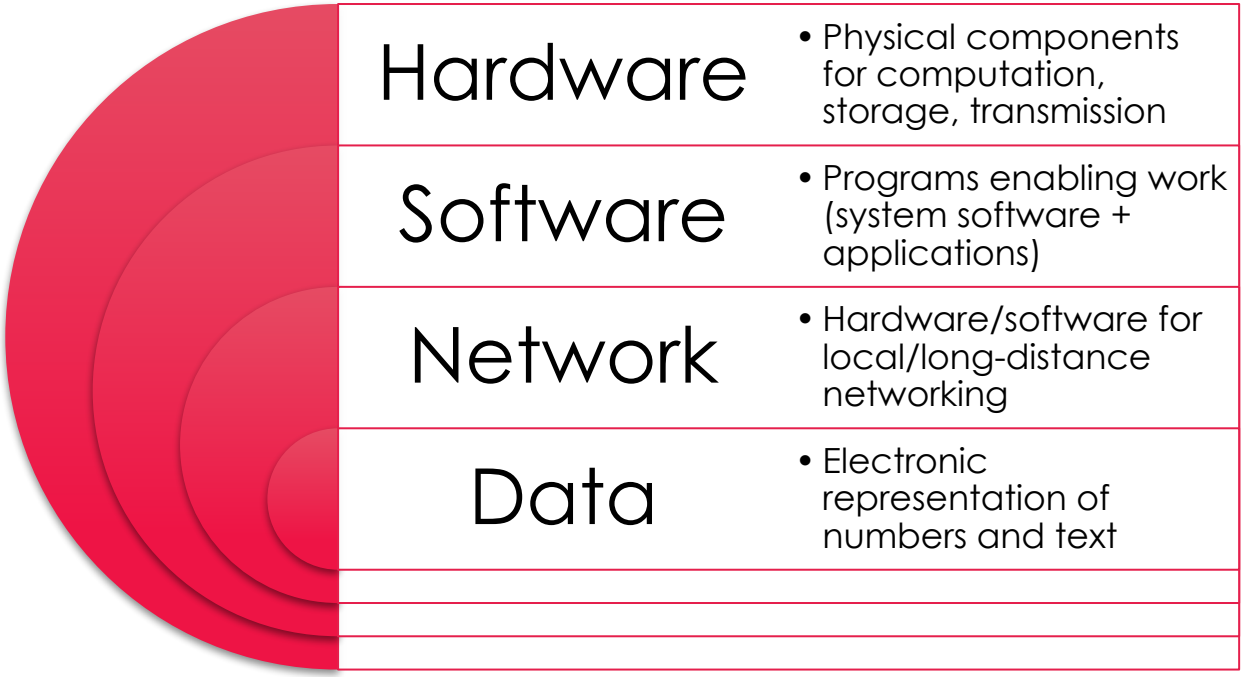


FIGURE 6.2 From strategy to architecture to infrastructure.

FRAMEWORK FOR INFRASTRUCTURE AND ARCHITECTURE ANALYSIS

- When developing a framework for transforming business strategy into architecture and then into infrastructure, these basic components should be considered:



FRAMEWORK FOR INFRASTRUCTURE AND ARCHITECTURE ANALYSIS

- These are example questions to ask for each basic component:

Component	What?	Who?	Where?
Hardware	What type of devices will users use?	Who knows most about servers?	Centralized or distributed?
Software	Does strategy require ERP software?	Who is affected by move to SAP?	Multiple database instances?
Network	How should network be structured?	Who needs network connection?	Lease cable or use satellite?
Data	What data needed for sales system?	Who needs access to sensitive data?	Cloud or data center?

FRAMEWORK FOR INFRASTRUCTURE AND ARCHITECTURE ANALYSIS

Hardware	Software	Network	Data
19 Servers: • 18 for sales • 1 for accounting LaCie 10-GB Thunderbolt RAID hard drive storage system	ERP system with modules for • Sales • Accounting • Inventory Enterprise application integration (EAI) software Apache operating system MySQL database software	• Cable modem to ISP • Dial-up lines for backup • Cisco routers, hubs, and switches • Firewalls from CheckPoint	Database • Sales • Inventory • Accounting • Complementary items

FIGURE 6.5 GiantCo.com's infrastructure components.

COMMON IT ARCHITECTURES

- **Centralized Architecture** - A large central computer system runs all applications and stores all data. All functionality is located in one place to make it easier to manage.
- **Decentralized Architecture** - Computing power is spread out among a number of devices in different locations. Typically chosen to modularize and address concerns about scalability.
- **Service-Oriented (Cloud-based) Architecture** - Software is broken down into services "orchestrated" and connected to each other in order to be agile, reusable, and facilitate rapid app design.
- **Software-Defined Architecture** - Infrastructure can be reconfigured autonomously based on rules to flexibly manage resources.

HOW DO ENTERPRISES DEVELOP THEIR MASSIVE SYSTEMS?



ENTERPRISE ARCHITECTURE

- A **comprehensive framework for creating the blueprint** for all information systems and their interrelationships in the business firm.
- The basis of enterprise architecture is business strategy.
- Specifies how information technologies support business processes, align with business needs, and produce business outcomes.
- Includes the standard technical capabilities and activities for all parts of the enterprise and guidelines for making choices.

ENTERPRISE ARCHITECTURE LAYERS

Business Layer	Application Layer	Information Layer	Technology Layer
• The common business organizations, strategies, functions, and models that describe how business is done and the important cross-functional business processes used in the business.	• The applications that support business processes, specifying relationships between processes and between applications.	• The enterprise information assets used by business processes and enterprise applications. This layer specifies physical storage and access for data such as operational and analytics data.	• The hardware and software infrastructure, including IT standards, structures, and the plans for IT, which create the technical foundation for the organization.

COMMON ENTERPRISE ARCHITECTURES

Architecture	Description	High-level Overview
TOGAF (the open group architecture framework)	A process for creating an enterprise architecture	Four architectures: business architecture, application architecture data architecture, and technical architecture
Zachman	A taxonomy for the artifacts of an enterprise architecture	Six descriptive foci for the architecture (data, function, network, people, time, and motivation) and six key player perspectives (planner, owner, designer, builder, subcontractor, and enterprise)
FEA (federal enterprise architecture)	Models, taxonomy, and processes originally created to build an enterprise architecture for the U.S. Government	Five reference models: business, service, components, technical, and data
Gartner	An active process for creating an enterprise architecture bringing together business owners, technology implementers, and information specialists	Uses an ongoing process of creating, maintaining, and revisiting the EA to keep it current and vital to the organization focused on the business strategy of where the business wants to go and how it will get there

FIGURE 6.7 Common enterprise architectures.

ACTIVITY



ACTIVITY (30 MINUTES)

- For each group, analyze the architecture and infrastructure of your current capstone system and answer the guide questions provided in the paper.
- Additionally, identify and describe the infrastructure components of your capstone system. Follow the example below:

Hardware	Software	Network	Data
19 Servers: • 18 for sales • 1 for accounting LaCie 10-GB Thunderbolt RAID hard drive storage system	ERP system with modules for • Sales • Accounting • Inventory Enterprise application integration (EAI) software Apache operating system MySQL database software	• Cable modem to ISP • Dial-up lines for backup • Cisco routers, hubs, and switches • Firewalls from CheckPoint	Database • Sales • Inventory • Accounting • Complementary items

FIGURE 6.5 GiantCo.com's infrastructure components.

THANK YOU FOR LISTENING!

Sources:

Bytheway, A. (2014). Investing in information: The Information Management Body of Knowledge. Springer.

Laudon, K. & Laudon, J. (2020). Management Information Systems